Ara Jeong

Assistant Professor, Department of Geography, Kyung Hee University

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RESEARCH INTERESTS

Quaternary Geochronology | Cosmogenic Radionuclides | Rock Coast Evolution | Geomorphic Hazards | Anthropocene Soil Erosion | Post-Wildfire Landscapes | UAV-SfM & LiDAR | Numerical Modeling

EDUCATION

2015–2019 **Ph.D., Geography**, Arizona State University, Tempe, AZ, USA
School of Geographical Sciences and Urban Planning
Dissertation: *Anthropocene in the geomorphology of the Sonoran Desert*
Committee: Ronald I. Dorn, Ian J. Walker, Mark Schmeeckle

2013–2015 **M.A., Geography**, Korea University, Seoul, Republic of Korea
Thesis: *Meteoric beryllium-10 applications for reconstructing ice shelf history: A case study from Larsen Ice Shelf-B, Antarctica*
Committee: Yeong Bae Seong, Gwang Hee Choi, Byung Yong Yu

2009–2013 **B.A., Geography Education**, Korea University, Seoul, Republic of Korea (with great honors)

2009–2013 **B.Sc., Earth and Environmental Sciences**, Korea University, Seoul, Republic of Korea (with great honors)

ACADEMIC APPOINTMENTS

2025.03–Present **Assistant Professor**, Department of Geography, Kyung Hee University

2022.02–2025.02 **Assistant Professor**, Department of Civil and Environmental Engineering, Korea Military Academy

2021.07–2021.12 **Full-time Researcher**, Institute of Geologic Hazard and Industrial Resources, Pusan National University

2020.07–2021.06 **Research Professor**, Institute of Future Land, Korea University

2020.01–2020.12 **Full-time Researcher**, Institute of Geologic Hazard and Industrial Resources, Pusan National University

2019.07–2020.01 **Postdoctoral Research Scholar and Lecturer**, Arizona State University, USA

GRADUATE POSITIONS

2015.08–2019.05 Teaching Assistant, School of Geographical Sciences and Urban Planning, Arizona State University

2013.02–2015.02 Research Assistant and Teaching Assistant, Department of Geography, Korea University

RESEARCH FUNDING

2026.03–2029.02 **Prediction of Collapse Threshold and Lifespan for Rocky Coasts in the Korean Peninsula by Environmental Setting**
National Research Foundation of Korea (NRF), Early Career Research Program (Type A) · Principal Investigator · KRW 245,679,000 · RS-2026-25474248

2025.03–2026.02 **Application of Numerical Dating Techniques for the Quantification of Landscape Processes, Forms, and History**
Kyung Hee University, Research Grant for New Faculty · Principal Investigator · KRW 30,000,000 · KHU-20251317

PUBLICATIONS

International Peer-Reviewed Journal Articles

16 **Jeong, A.**, Seong, Y. B., Weber, J., & Yu, B. Y. (submitted). Revisiting glacial and paraglacial landscape in the Beartooth Mountains, southern Montana: An integrated chronology of moraines and terraces. *Annals of the American Association of Geographers*.

15 **Jeong, A.**, Seong, Y. B., Choi, K. H., & Swirad, Z. M. (2026). Global patterns of shore platform evolution and cliff retreat: Insights from cosmogenic radionuclides. *Earth-Science Reviews*, 272, 105323. doi:10.1016/j.earscirev.2025.105323

14 **Jeong, A.**, Connor, D. S., Dorn, R. I., & Seong, Y. B. (2025). Estimating urbanization's impact on soil erosion: A global comparative analysis and case study of Phoenix, USA. *Land*, 14(8), 1590. doi:10.3390/land14081590

13 **Jeong, A.**, Seong, Y. B., Choi, K. H., Swirad, Z. M., Lee, C. H., & Yu, B. Y. (2024). Shore platform erosion and cliff retreat in the Eastern Korea: A quantified assessment using ^{10}Be concentrations and numerical modeling. *Marine Geology*, 472, 107291. doi:10.1016/j.margeo.2024.107291

12 **Jeong, A.**, Seong, Y. B., Dorn, R. I., & Yu, B. Y. (2024). Precipitation as a key control on erosion rates in the tectonically inactive northeastern Sonoran Desert, central Arizona, USA. *Physical Geography*, 45(1), 53–83. doi:10.1080/02723646.2023.2251654

11 **Jeong, A.**, Seong, Y. B., Gootee, B. F., Yu, B. Y., & Cheung, S. Y. (2023). Application and limitations of the authigenic ^{10}Be/^{9}Be dating and meteoric ^{10}Be inventory in Bouse Formation along the lower Colorado River corridor, southwestern USA. *Episodes*, 46(1), 85–97. doi:10.18814/epiiugs/2022/022018

10 Rhee, H. H., Seong, Y. B., Lee, M. K., **Jeong, A.**, Dash, C., Lee, J. I., Yoo, K. C., & Yu, B. Y. (2022). Spatial variations of authigenic beryllium isotopes in surface sediments of the Antarctic oceans: A proxy for sea ice dynamics and sedimentary environments. *Geosciences Journal*, 26(4), 455–467. doi:10.1007/s12303-022-0003-4

9 **Jeong, A.**, Dorn, R. I., Seong, Y. B., & Yu, B. Y. (2021). Acceleration of soil erosion by different land uses in arid lands above ^{10}Be natural background rates: Case study in the Sonoran Desert, USA. *Land*, 10(8), 834. doi:10.3390/land10080834

8 **Jeong, A.**, Seong, Y. B., Lee, C. H., & Choi, K. H. (2021). A numerical model study for simulation of rocky coast evolution and erosion using cosmogenic nuclides: A case study along the Dunduri and Dokdo shore platform in Korea. *Journal of the Geological Society of Korea*, 57(2), 195–207. doi:10.14770/jgsk.2021.57.2.195

7 Skotnicki, S. J., Seong, Y. B., Dorn, R. I., Larson, P. H., DePonty, J., & **Jeong, A.** (2021). Drainage integration of the Salt and Verde rivers in a Basin and Range extensional landscape, central Arizona, USA. *Geomorphology*, 374, 107512. doi:10.1016/j.geomorph.2020.107512

6 Larson, P. H., Dorn, R. I., Skotnicki, S. J., Seong, Y. B., **Jeong, A.**, & DePonty, J. (2020). Impact of drainage integration on basin geomorphology and landform evolution: A case study along the Salt and Verde rivers, Sonoran Desert, USA. *Geomorphology*, 371, 107439. doi:10.1016/j.geomorph.2020.107439

5 Douglass, J. C., Gootee, B. F., Dallegge, T., **Jeong, A.**, Seong, Y. B., & Yu, B. Y. (2020). Evidence for the overflow origin of the Grand Canyon. *Geomorphology*, 369, 107361. doi:10.1016/j.geomorph.2020.107361

4 **Jeong, A.** (2019). Sediment accumulation expectations for growing desert cities: A realistic desired outcome to be used in constructing appropriately sized sediment storage of flood control structures. *Environmental Research Letters*, 14(12), 125005. doi:10.1088/1748-9326/ab30e5

3 **Jeong, A.**, & Dorn, R. I. (2019). Soil erosion from urbanization processes in the Sonoran Desert, Arizona, USA. *Land Degradation & Development*, 30(2), 226–238. (cover article) doi:10.1002/ldr.3207

2 **Jeong, A.**, Lee, J. I., Seong, Y. B., Balco, G., Yoo, K. C., Yoon, H. I., Domack, E., Rhee, H. H., & Yu, B. Y. (2018). Late Quaternary deglacial history across the Larsen B embayment, Antarctica. *Quaternary Science Reviews*, 189, 134–148. doi:10.1016/j.quascirev.2018.04.011

1 Dorn, R. I., & **Jeong, A.** (2018). Rock coating and weathering-rind development at the edge of retreating glaciers: An initial study. *Yearbook of the Association of Pacific Coast Geographers*, 80, 66–96. doi:10.1353/pcg.2018.0004

Domestic Peer-Reviewed Journal Articles (Korea)

3 **Jeong, A.** (2020). Spatial analysis for detrital zircon provenance study: An initial study. *Journal of the Korean Geographical Society*, 55(4), 439–451. doi:10.22776/kgs.2020.55.4.439

2 **Jeong, A.** (2020). Understanding the transport mechanism of paleo-deposits in the Sonoran Desert with grain size distribution and shape analysis. *Journal of the Association of Korean Geographers*, 9(2), 265–279. doi:10.25202/JAKG.9.2.3

1 **Jeong, A.** (2019). Human impacts on urban landscapes in North American desert: A case study in the Phoenix, Arizona, USA. *Journal of the Korean Geomorphological Association*, 26(3), 69–85. doi:10.16968/JKGA.26.3.69

Book Chapters

3 Dorn, R. I., Larson, P. H., & **Jeong, A.** (2025). Landforms on faulted rocks. In R. Schaetzl & R. Marston (Eds.), *Essentials of Geomorphology* (Ch. 10, pp. 182–211). Cambridge University Press. doi:10.1017/9781108781459

2 Dorn, R. I., Gordon, S. J., & **Jeong, A.** (2022). Nanoscale mineral decay and its importance in geomorphology. In J. F. Shroder (Ed.), *Treatise on Geomorphology* (2nd ed., Ch. 3.09, pp. 116–150). Academic Press. doi:10.1016/B978-0-12-818234-5.00150-4

1 **Jeong, A.**, Cheung, S. Y., Walker, I. J., & Dorn, R. I. (2018). Urban geomorphology of an arid city: Case study of Phoenix, Arizona. In M. J. Thornbush & C. D. Allen (Eds.), *Urban Geomorphology: Landforms and Processes in Cities* (Ch. 10, pp. 177–204). Elsevier. doi:10.1016/B978-0-12-811951-8.00010-2

PRESENTATIONS

Invited Talks

2024 Anthropocene: Humans and nature at the crossroads of coexistence. 87th Summer Workshop, Academic Communication Society, Korea University, Seoul. (in Korean)

2014 Late Quaternary deglacial history across the Larsen B embayment, Antarctica. 20th International Symposium on Polar Sciences, Incheon.

Conference Presentations

2026 **Jeong, A.**, Seong, Y. B., & Choi, K. H. Quantifying millennial-scale coastal erosion and shore platform evolution using cosmogenic radionuclides and numerical models. 10th International Conference on Asian Marine Geology (ICAMG-10), Busan. (Oral)

2026 **Jeong, A.**, Seong, Y. B., & Choi, K. H. Synthesis of shore platform research in Korea and future directions. Winter Conference, Korean Geomorphological Association. (Oral, in Korean)

2026 Seong, Y. B., **Jeong, A.**, & Lee, G. Moraines and terraces of the Beartooth Mountains, Montana, USA: Geochronologic constraints on their paraglacial linkage. Winter Conference, Korean Geomorphological Association. (Oral)

2025 **Jeong, A.**, Seong, Y. B., & Choi, K. H. Evolution of rocky coasts in Korea: Effects of relative sea-level change revealed by cosmogenic radionuclides. Summer Conference, Korean Geomorphological Association. (Oral, in Korean)

2024 **Jeong, A.** Quantitative assessment of shore platform erosion and cliff retreat on the East Coast of Korea using ^{10}Be and numerical modeling. Annual Conference, Korean Geographical Society. (Oral, in Korean)

2023 **Jeong, A.** Factors controlling long-term catchment-averaged erosion rates in central Arizona, USA: Climate, tectonics, and lithology. Annual Conference, Korean Geographical Society. (Oral, in Korean)

2022 **Jeong, A.**, Seong, Y. B., Gootee, B. F., Yu, B. Y., & Cheung, S. Y. The potential and limitations of the meteoric ^{10}Be dating technique on ancient finely laminated lake deposits. Summer Conference, Korean Geomorphological Association. (Oral, in Korean)

2019 **Jeong, A.** Acceleration of soil erosion from urban sprawl. American Association of Geographers Annual Meeting, Washington, DC, USA.

2018 **Jeong, A.** Land degradation associated with urban expansion in a warm desert setting of metropolitan Phoenix, Sonoran Desert, western USA. Binghamton Geomorphology Symposium, Syracuse, NY, USA.

2014 **Jeong, A.** Late Quaternary deglacial history across the Larsen B embayment, Antarctica. Annual Meeting, Korean Geomorphological Association, Seoul.

Professional and Public Lectures

2025 Understanding the Anthropocene: Its concept, characteristics, and the impact of human activities on the Earth system. 3rd Science Concert, College of Sciences, Kyung Hee University. (in Korean)

2025 Anthropocene: Humans and nature at the crossroads of coexistence — An age of crisis or an age of possibility? "Beyond the Wall" lecture series, Kyung Hee University. (in Korean)

2024 Anthropocene: Humans and nature at the crossroads of coexistence — Geography in an age of crisis or an age of possibility? Special Lecture, Kyung Hee University. (in Korean)

2019 Acceleration of soil erosion from urban sprawl. JE Fuller Hydrology & Geomorphology, Inc., Tempe, AZ, USA.

TEACHING

Kyung Hee University · 2025–Present

GRADUATE
- General Geomorphology (GEOG702000)
- Climate Science (CSSC700400)
- Geographic Perspective (CSSC700200)

UNDERGRADUATE
- General Geomorphology (GEOG200800)
- Natural Hazard (GEOG302900)
- Independent Learning & Research 2 (GEOG305900)
- From Big Bang to Civilizations (GEC0103S19, core)

Korea Military Academy · 2022–2025
- Military Geography and Meteorology
- Earth Environmental Systems
- Environmental Information Systems
- Global Environmental Change (general education)

Korea University · Lecturer, 2020–2022
- Geomorphology (Spring 2022)
- Nature and Human Life (general education, Fall 2020)

Arizona State University · 2015–2019

- Instructor, GPH 211 Introduction to Landform Processes (2016–2019)
- Instructor, GPH 111/112 Introduction to Physical Geography (2018)
- Teaching Associate, GPH 111 Physical Geography Lab (2015–2017)

Korea University · Teaching Assistant, 2013–2014

- GEOG 371 Research and Survey Techniques in Geography (2014)
- GEOG 201 Geomorphology (2014)
- GEOG 152 Introduction to Physical Geography (2013)

STUDENT SUPERVISION

2026–Present	Eungi Cho, M.S. student, Kyung Hee University
2023	Cadet poster, Korea Military Academy: "Analysis of heavy-metal contamination sites using ArcGIS and recommendations for environmental regulations." Annual Conference, Korean Geographical Society. (in Korean)

PROFESSIONAL SERVICE

2024–Present	Board Member, The Korean Geomorphological Association
Reviewer	*Geomorphology*, *Earth Surface Dynamics*, *Land Degradation & Development*, *Sediment Geology*, *Episodes*, *Land*, *Journal of the Korean Geographical Society*, *Journal of the Korean Geomorphological Association*, *Journal of the Association of Korean Geographers*, *Journal of the Korean Earth Science Society*, *Economic and Environmental Geology*
Memberships	The Korean Geographical Society · The Korean Geomorphological Association · Korea Association for Quaternary Research

FELLOWSHIPS, AWARDS, AND HONORS

2019	Travel Award, School of Geographical Sciences and Urban Planning, Arizona State University
2019	Travel Award, Graduate and Professional Student Association, Arizona State University
2018	Binghamton Geomorphology Symposium Award (NSF Award No. 1832278)
2018	JE Fuller Fellowship, JE Fuller Hydrology & Geomorphology, Inc.
2014	National Graduate Research Fellowship, Ministry of Education, Republic of Korea
2012	Global Human Resource Development Fellowship, Korea University
2010–2012	Learning Guide Scholarship, Government of the Republic of Korea
2010–2011	Best Honors Fellowship (2010) and Honors Fellowship (2011), Korea University
2010–2011	Study Scholarship, Korea University